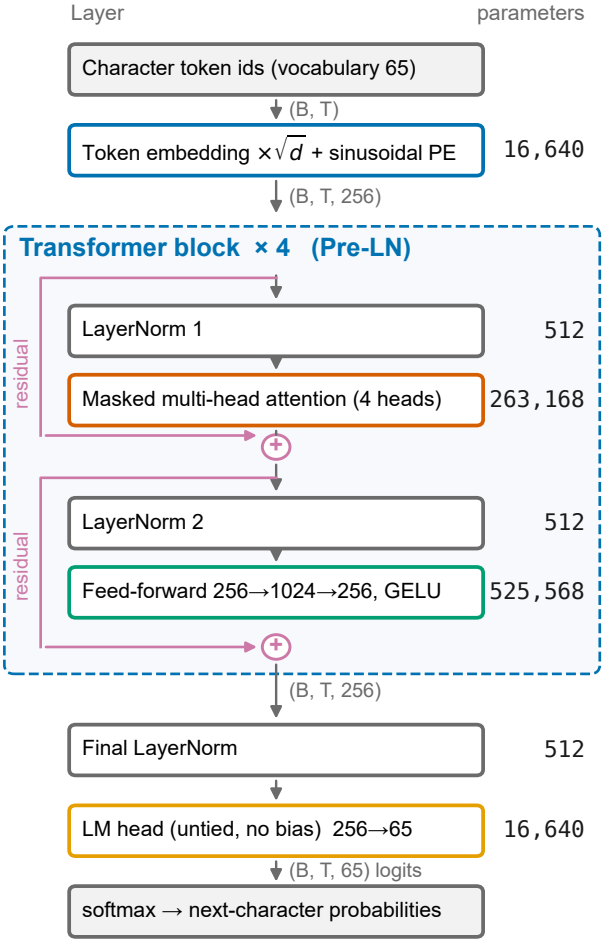


Parameter budget

Token embedding (0.5%)	16,640
4 × attention (33.0%)	1,052,672
4 × feed-forward (65.8%)	2,102,272
4 × 2 LayerNorms (0.1%)	4,096
Final LayerNorm (0.02%)	512
LM head (0.5%)	16,640
Total	**3,192,832**

d_model = 256
heads = 4 (d_h = 64)
layers = 4
context T = 128
FFN width = 1024
dropout = 0.1
vocabulary = 65